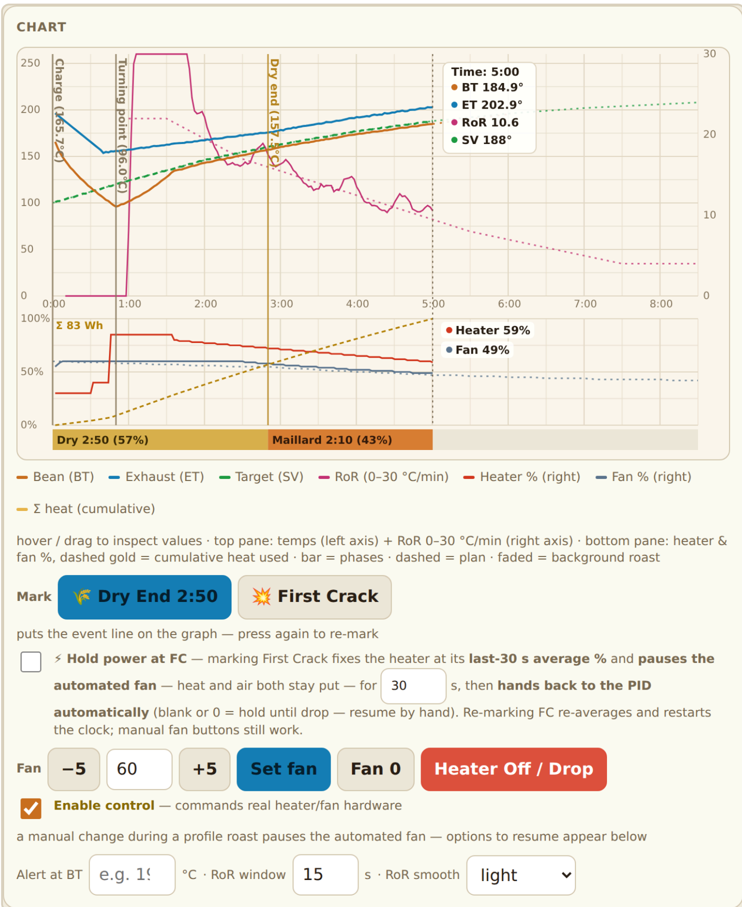
🔥 Roast Controller User Manual

Web-Bluetooth controller for Hamid / Matchbox (TC4-class) coffee roasters



Covers setup, your first roast, the everyday roasting loop, profile design, inventory, cloud sync and troubleshooting. Screenshots show the app in day mode. Edition of September 2026.

Contents

> **Read this first.** The app drives a real heating element. Controls stay locked until you tick **Enable control**. Keep the fan running whenever the heater is on, never leave a roast unattended, and press `Heater Off / Drop` (the red button under the chart) if anything looks wrong. It always works, whatever else is running.

1. Before you start

You need

- **Chrome or Edge** on Android, Windows, macOS, Linux or ChromeOS, with Bluetooth switched on. iOS Safari does not support Web Bluetooth, so an iPhone or iPad will not work. A desktop or an Android phone or tablet is the usual choice.
- The app opened over **HTTPS** (the hosted link you were given). Opening the file from disk will not work because Web Bluetooth needs a secure page.
- A roaster whose Bluetooth module advertises as **MATCHBOX** or **MBOX**, powered on and within a few metres.
- Green coffee, a scale, and somewhere ventilated to roast.

Good to know

- Everything you save (profiles, roast logs, inventory, settings) is stored in **this browser on this device**. Nothing leaves your machine unless you switch on Cloud sync (section 10).
- The ☀️ / 🌙 button in the header switches day and night mode. The choice is remembered.
- The page is one long column of panels. This manual follows them from top to bottom.

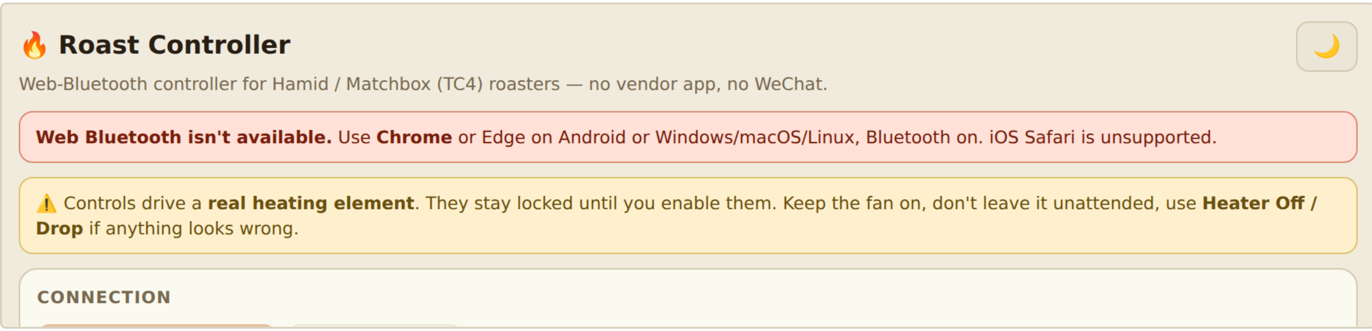


Fig. 1 — The header with the day/night toggle, and the safety banner shown on every visit.

2. Getting to know the screen

The whole app is one page. Scroll down and you pass through these panels in order:

1. **Connection** — connect or disconnect the roaster, see the Bluetooth details.
2. **Live** — roast timer, batch code, the six live tiles (Bean, ET, RoR, Target, Heater, Fan), the previous event and live phase bar.
3. **Chart** — temperatures, RoR, heater and fan over time, the Mark buttons, the fan row with `Heater Off / Drop` and **Enable control**, alerts and RoR settings.
4. **Between batches / charge** — cool-to, charge temperature and fan, the charge soak, Auto-charge, and the `Start between-batch cycle` button.
5. **Profile designer** — pick, generate, draw, import, export and run roast profiles.
6. **Advanced mode** tick — reveals the ET-track and Power profile tools (section 11).
7. **Data** — CSV export, saved roasts, roast notes and stars, background overlay, raw log.
8. **Cloud sync** — share your data between devices through your own Cloudflare Worker.
9. **Green coffee inventory** — beans in stock, label photo reader, batch planner.

Fig. 2 — The full page at a glance (day mode). The Live rail sits beside the chart on wide screens and above it on phones.

3. First-time setup (5 minutes)

1. Power the roaster on. Close the vendor app if it is open. Only one thing can hold the Bluetooth connection.
2. Open the app and press `Connect roaster` in the **Connection** panel. A browser dialog lists nearby devices. Pick `MATCHBOX` or `MBOX`. If it is not listed, tick **Can't find it by name? List all Bluetooth devices** and try again.
3. The status pill turns to **connected** and the **Live** tiles start showing Bean °C, ET °C, Heater % and Fan %. If the numbers appear, the link is working.
4. Tick **Enable control** in the fan row under the chart. Until this is ticked the app only *reads* from the roaster.
5. Test the fan: type `50` next to `Set fan` and press it. You should hear the roaster's fan change and see Fan % update. This confirms commands reach the machine.

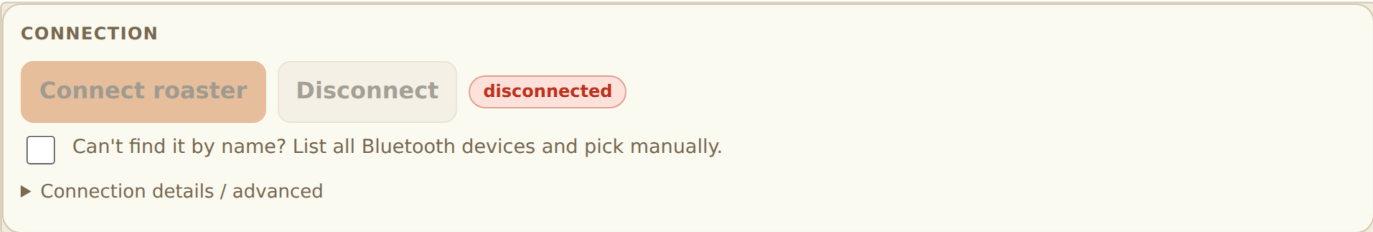


Fig. 3 — The Connection panel before connecting.

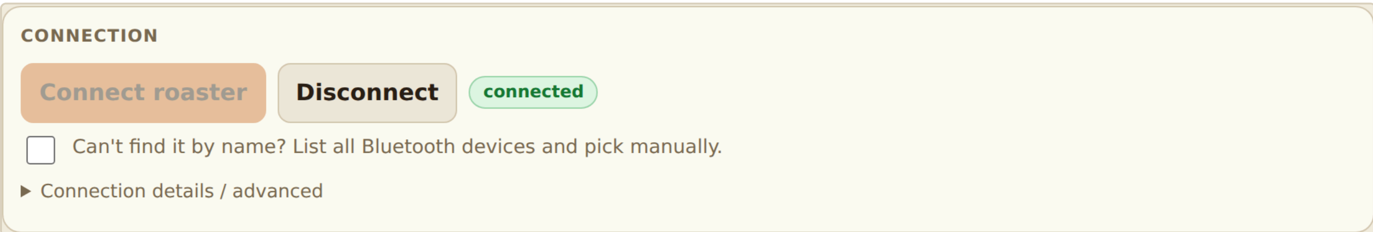


Fig. 4 — Connected. The pill turns green and Disconnect becomes available.

> If the tiles stay at – after connecting, open **Connection details / advanced** and see the troubleshooting table in section 13.

4. Your first roast, step by step

The app can run the *whole* roast for you from a single button. For the first roast, use a ready-made profile so you can learn the screen without designing a curve.

A. Load a profile

1. Download one of the ready-made curves from the repository's `profiles/` folder. `sprint-140-filter-208.json` is a good all-round filter roast for about 150 g.
2. In the **Profile designer** panel press `Import JSON` and choose the file. The curve appears on the graph with the name and notes filled in.
3. Press `Save profile`. It now appears in the profile drop-down for next time.
4. Check that **Auto fan** below the graph is ticked (the sample profiles set it). Leave the rest as is.

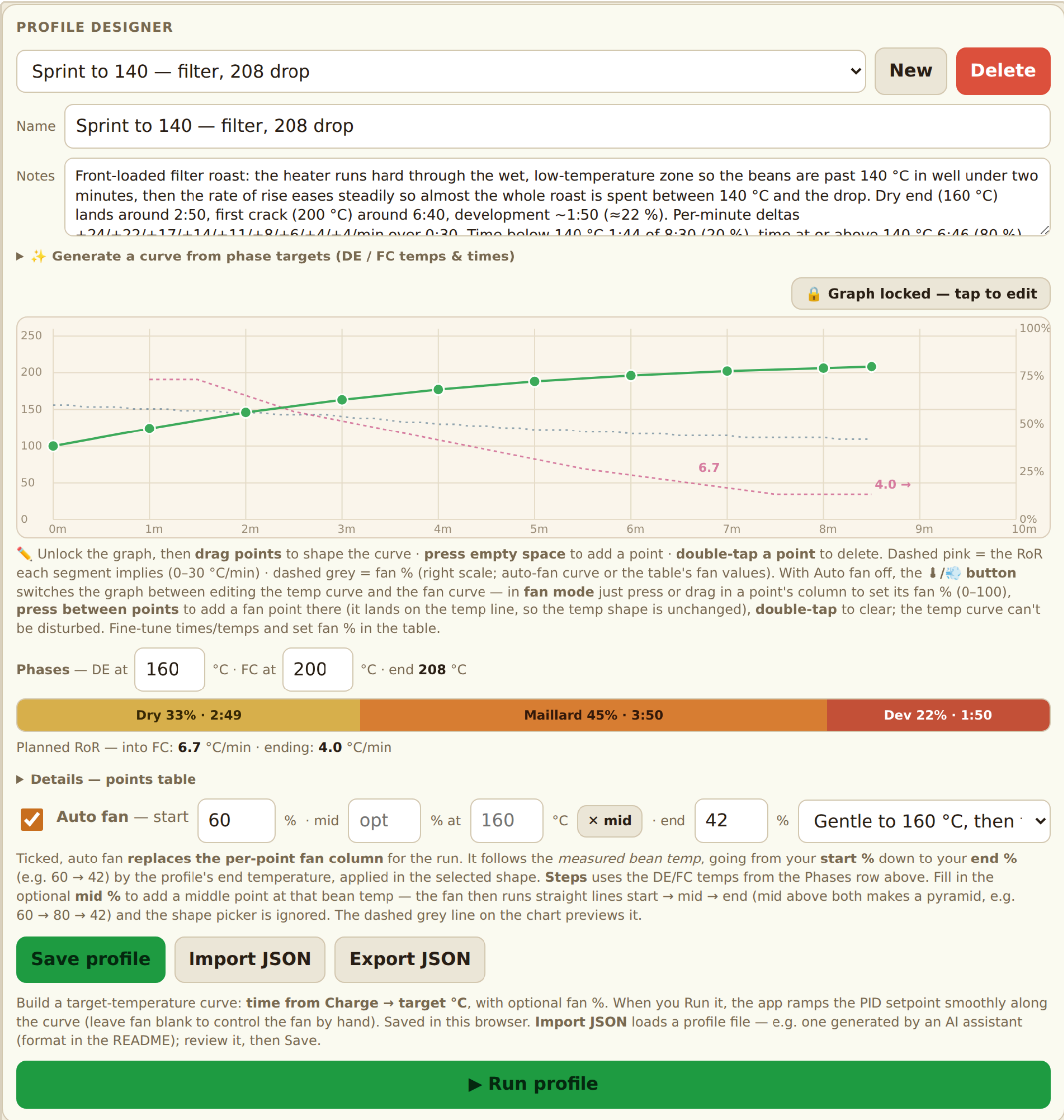


Fig. 5 — The sample profile loaded and saved. Green line: target bean temperature. Dashed pink: the RoR each segment implies. Dashed grey: the auto-fan curve. The phase bar shows the Dry / Maillard / Development split.

B. Set the between-batch cycle

In the **Between batches / charge** panel:

- **Cool to °C** — the machine is fanned down to this before charging (default 100).
- **Charge temp °C** — the drum temperature at which you drop beans in (default 180). Start with the default.
- **Charge fan %** — fan speed used while heating to charge and at the moment you pour (default 30).
- Leave **Charge soak** and **Auto-charge** ticked. Under **At bean drop** leave **Run the profile** selected. The preview box confirms which profile will run when the beans go in.

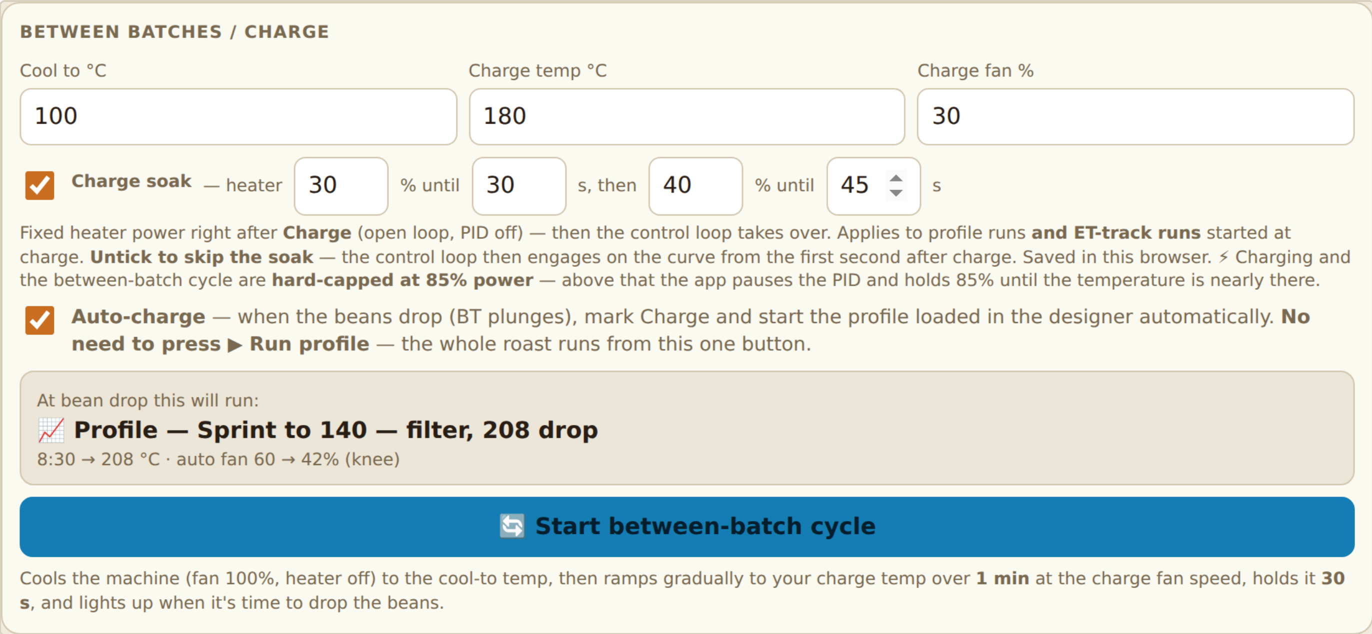


Fig. 6 — Between batches / charge, with the defaults. The grey box previews what will run at bean drop.

C. Run the cycle

1. Press 🔄 Start between-batch cycle. A dialog asks for the **bean** (pick one from inventory, or leave it untracked) and the **green weight** (150 g, 100 g, or a custom figure). Both are stored with the roast, and the green weight is taken out of that bean's stock at Drop. Press ▶ Start cycle.
2. The app fans the machine down to the cool-to temperature, then ramps to the charge temperature over about a minute and holds it for 30 s.
3. When the drum is stable the panel flashes ⬇ **DROP BEANS NOW** and the device beeps or vibrates. Pour the beans in.
4. With **Auto-charge** on, the app sees the bean-probe plunge, marks **Charge**, starts the timer, and begins the profile. You do not need to press Run profile.
5. For the first 45 s the heater runs at a fixed low power (the *charge soak*), then the PID takes over and follows your curve.

Fig. 7 — The New batch dialog: choose the bean and the green weight, then Start cycle.

D. During the roast

- Watch **Bean °C** climb and the **RoR** tile settle. On the chart the plan is dashed and the actual bean temperature is solid.
- Press 🌾 Dry End when the beans turn from green-yellow to tan (around 150–165 °C bean temperature) and 💥 First Crack when you hear the first pops. These marks drive the phase bar and the development-time percentage.
- If the beans are not tumbling well, bump the fan with +5. This pauses the automated fan. Resume profile fan hands it back.

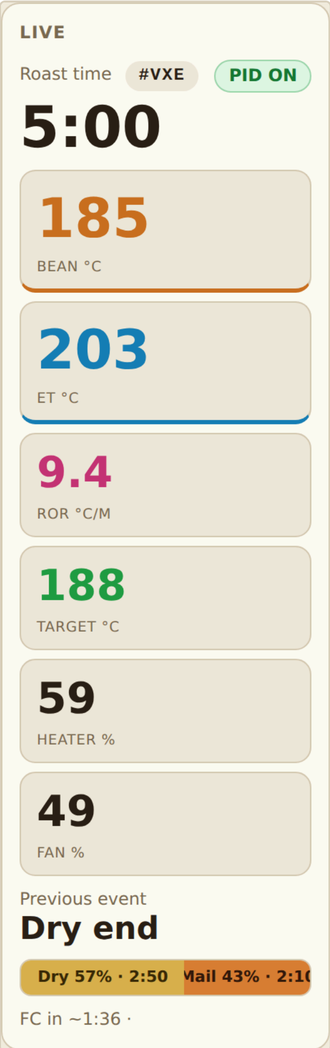


Fig. 8 — The Live rail at 5:00: timer, batch code, tiles, previous event, phase bar and the FC countdown.

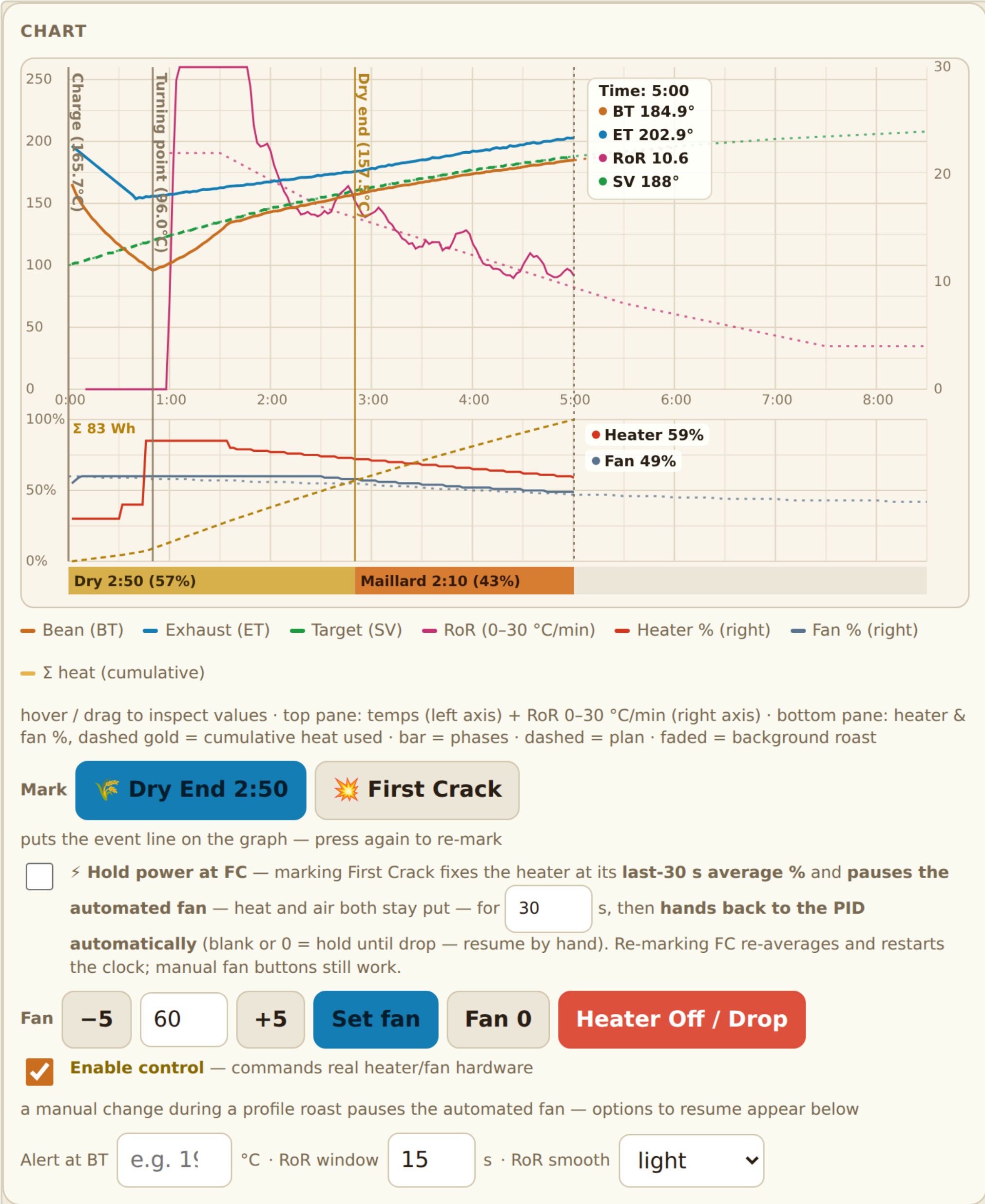


Fig. 9 — The chart at 5:00 with Charge, turning point and Dry End marked. Hover or drag to read values. Below the chart: Mark buttons, FC power hold, the fan row with Heater Off / Drop, and the alert and RoR settings.

E. Drop

1. When the beans reach the colour you want, press `Heater Off / Drop`. The heater goes to 0, the fan returns to the charge speed, the roast is stamped with a 3-character **batch code** (for example `#K7Q`) and is **saved automatically**. Write the code on the bag.
2. If you let the profile run to its last point instead, the heater switches off by itself and an alert sounds. You still press `Heater Off / Drop` to end and save the roast.
3. The saved roast appears at the top of the saved-roasts list in the **Data** panel, named by its batch code, time and bean. Type tasting notes there whenever you like.
4. Tip the beans out to cool and let the machine idle with the fan on before the next batch.

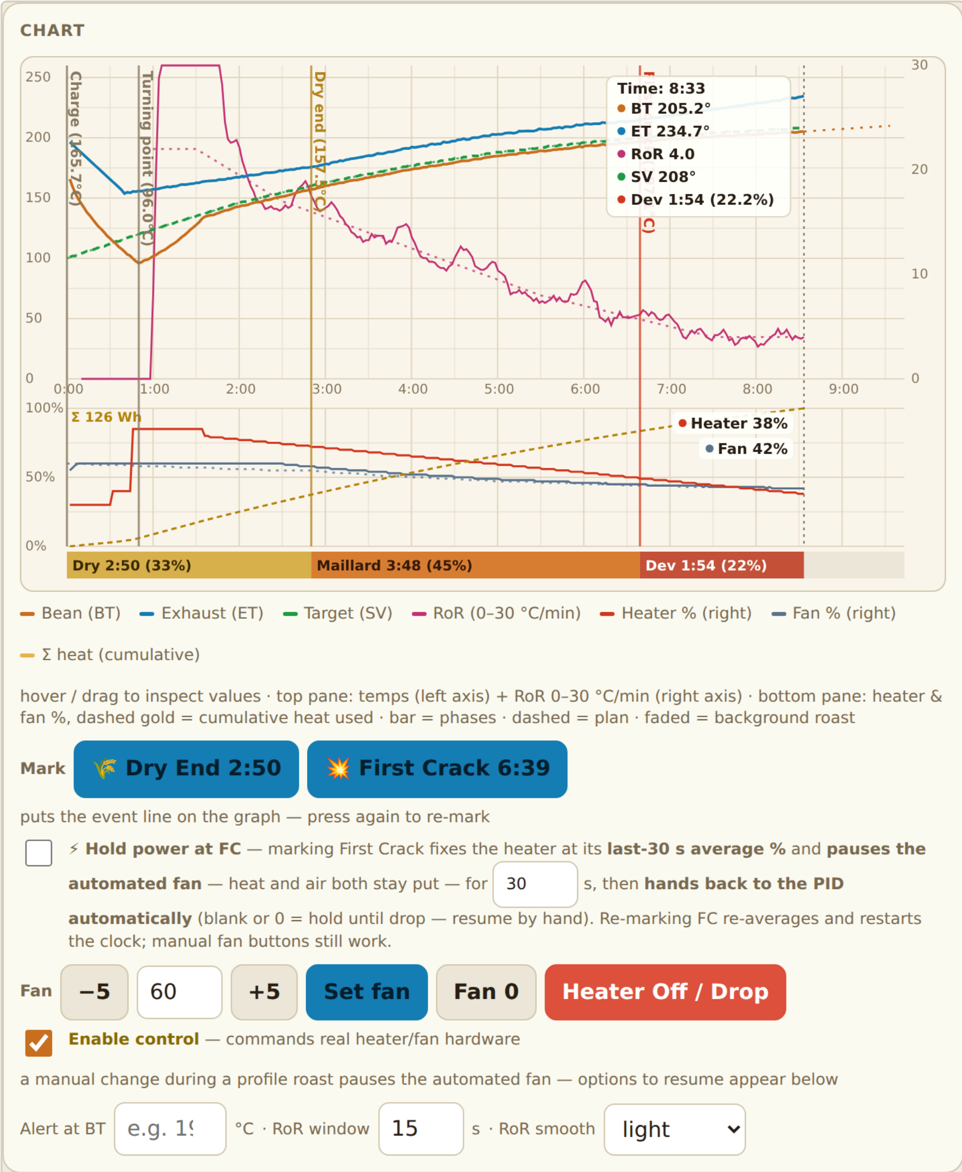


Fig. 10 — Just before Drop at 8:30. First Crack is marked and the development phase is shown on the bar.

That is the whole loop. The next batch starts again at step C.

5. The everyday workflow

```
Connect → Enable control → pick profile → Start between-batch cycle
   → DROP BEANS NOW → (auto-charge) → mark Dry End / First Crack
   → Heater Off / Drop → notes / star → next batch
```

Between batches the app does the cooling and re-heating for you, so the rhythm is: *tip out, note, press Start cycle, wait for the light, pour*.

Tips for a smooth session

- Set the profile and the between-batch values once at the start. They are remembered.
- Keep the bag or a notebook beside you: the batch code is shown in the Live rail from Charge onward.
- Use **Overlay as background** (section 8) with a roast you liked, and roast the next batch to match it.
- Green weight is deducted from inventory at Drop, so keep your stock up to date and the batch planner will keep suggesting even splits.

6. Building your own profile

Open the **Profile designer** panel. A profile is a target bean-temperature curve over time (from Charge), plus an optional fan schedule.

Fastest: generate from phase targets

1. Open ✨ **Generate a curve from phase targets**.
2. Enter the temperature you want at 0:00, your **dry-end** temperature and how long drying should take, your **first-crack** temperature and how long the Maillard phase should take, and the **drop** temperature and development time.
3. Pick a **shape**. If unsure, start with **Declining RoR** (the classic). The others are explained in the panel text. **Sprint** and **Flat development** are popular next steps.
4. Optionally tick ⚡ **Precharge** (extra push in the first 40 s) or 🍃 **Coast** (finish 3 °C early on stored heat).
5. The curve loads into the graph with the name, notes and phase markers filled in. Flip between shapes to compare, then Save profile.

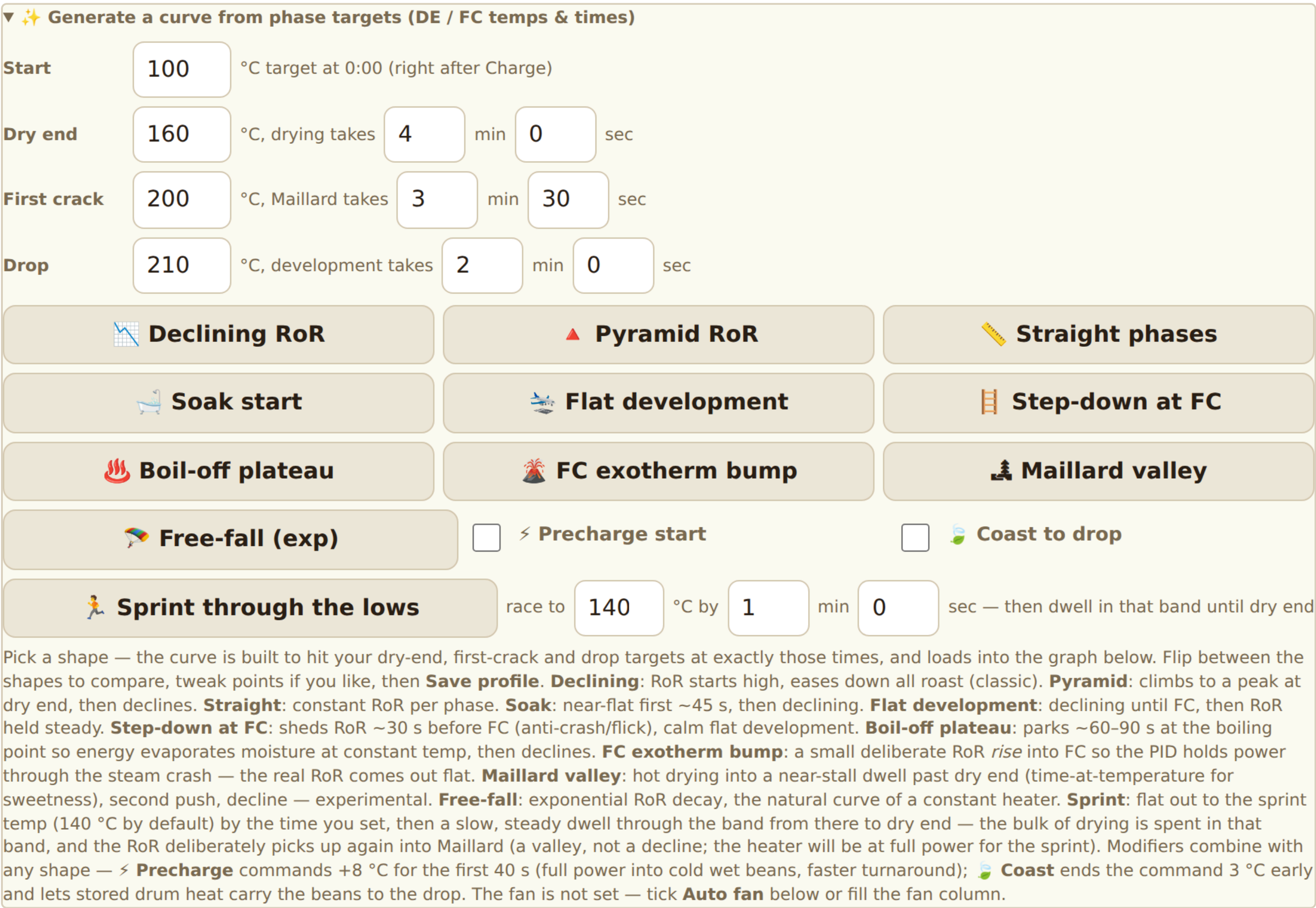


Fig. 11 — The generator: phase temperatures and durations, the shape picker, and the two modifiers.

Hands-on: draw on the graph

- Press 🔒 Graph locked — tap to edit to unlock.
- Drag points to move them, press empty space to add a point, double-tap a point to delete it. The implied RoR of each segment is shown while you edit.

- The **phase bar** uses the **DE at** and **FC at** temperatures beside it. Set those to match your beans.
- Fine-tune numbers in **Details — points table**.

▼ **Details — points table**

TIME M:SS	TARGET °C	FAN % (OPT)	
0:00	100	opt	×
1:00	124	opt	×
2:00	146	opt	×
3:00	163	opt	×
4:00	177	opt	×
5:00	188	opt	×
6:00	196	opt	×
7:00	202	opt	×
8:00	206	opt	×
8:30	208	opt	×

Fig. 12 — The points table: time from Charge, target °C and an optional per-point fan %.

Fan

- **Auto fan** (recommended): the fan follows the *measured* bean temperature, declining from the start % to the end %. An optional **mid** point lets you shape a rise-then-fall. This overrides per-point fan values while running.
- Or fill the **Fan %** column per point for explicit control.
- Or leave both empty to run the fan by hand during the roast.

Import and export

`Export JSON` saves the profile as a file you can share or hand to an AI assistant to tweak. `Import JSON` loads one. The format is documented in `PROFILE_FORMAT.md` in the repository. Paste that file into Claude or another assistant to have it design a curve for a specific bean.

Running a profile without the between-batch cycle

Select it in the drop-down and press `▶ Run profile`. If Charge was marked within the last 10 seconds the charge soak applies. Otherwise the PID engages on the curve immediately, which is useful for restarting mid-roast.

7. Reading the screen during a roast

Live tiles

Bean (BT), Exhaust (ET), RoR (rate of rise in °C/min), Target (the setpoint the app is commanding), Heater % and Fan %. The **PID** pill shows whether the roaster's closed-loop mode is engaged. Below the tiles: the previous event, a live phase bar, and the estimated time to first crack.

Chart

Top pane: temperatures on the left axis, RoR on the right (0–30 °C/min). Bottom pane: heater and fan %, with the dashed gold line showing cumulative heat. Dashed curves are the plan. Faded curves are a background roast (section 8). Hover or drag on the chart to read values.

Under the chart

- **Mark** buttons for Dry End and First Crack. Press again to re-mark.
- ⚡ **Hold power at FC** — when you mark First Crack, freezes the heater at its recent average and pauses the automated fan for a set number of seconds, then returns to the PID. Useful to ride through the first-crack crash. Off by default.
- **Fan row** — quick ±5 nudges, `Set fan`, `Fan 0`, `Heater Off / Drop`, and the **Enable control** tick.
- **Alert at BT** — beeps when the bean temperature reaches this value. **RoR window** and **RoR smooth** tune how responsive versus how calm the RoR trace is.

Control rail

On wide screens a **Controls** column sits to the right of the chart (on phones it appears below it), laid out like the stock app's side panel: a **Target temp** tile and a **PID** tile, **Heater −/+** and **Fan −/+** steppers showing the live values, a green **event-tagging** button that tags **DE** then **FC**, and a red **Stop / Drop**. Both steppers move 1 % per tap (the fan row under the chart steps by 5 %). Heater ± takes **manual control**: it switches the PID off and stops any running profile or cycle (you are asked to confirm), then sets the heater directly. Stop is the same as `Heater Off / Drop`.

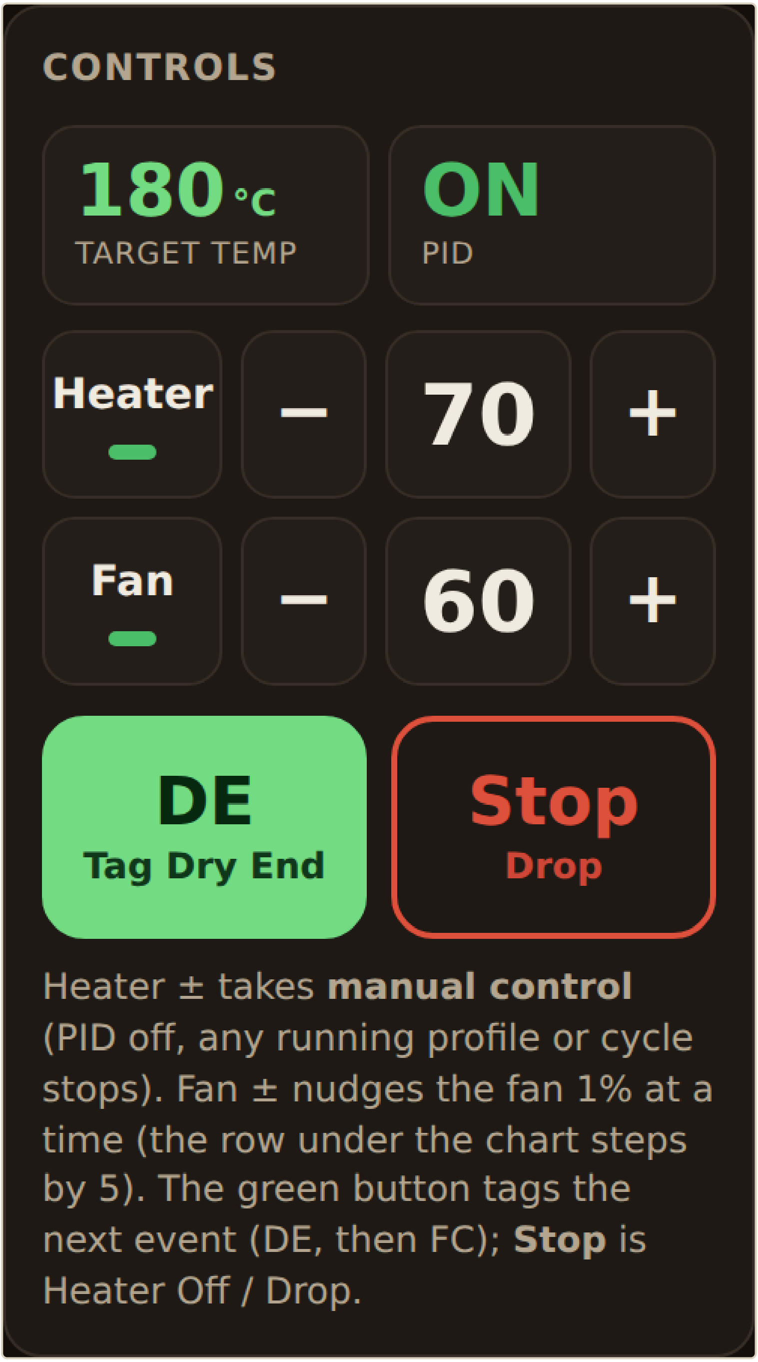


Fig. 13a — The control rail during a roast: the next event to tag is Dry End.

Manual mode

If you would rather roast by hand than follow a profile, choose 🎛 **Manual** under **At bean drop** in the **Between batches / charge** panel. The cycle cools, charges and lights **DROP BEANS NOW** exactly as before. When the beans go in, the app marks Charge, starts the timer and batch code, runs the charge soak if it is on, then sets your **opening heater %** and **fan %** with the PID **off** and hands over. From there you drive the roast with the Heater and Fan ± steppers, tag DE and FC with the green button, and press Stop / Drop to end and save. The opening values are remembered between sessions.

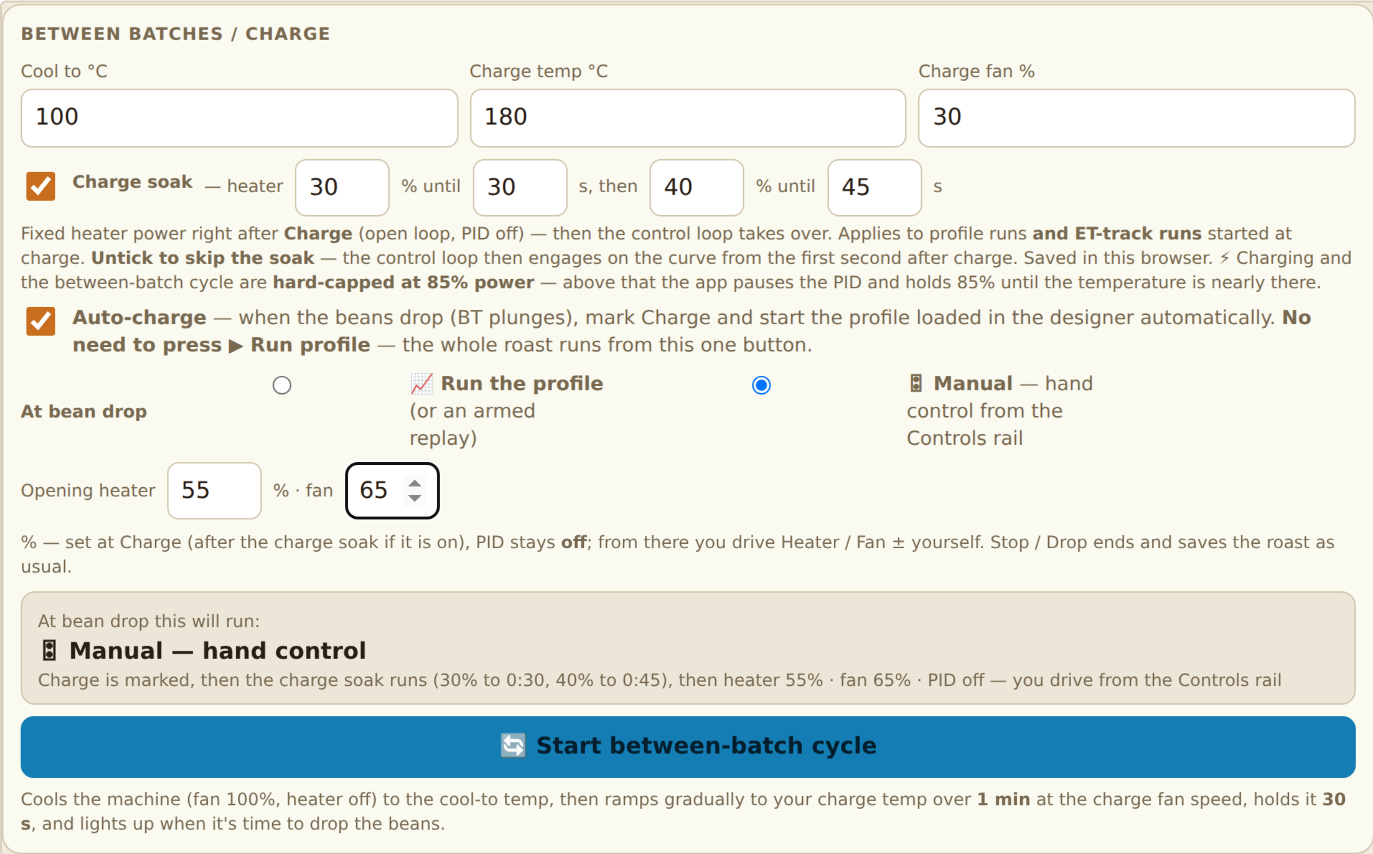


Fig. 13b — Manual mode selected: opening heater 55 %, fan 65 %. The preview box spells out what will happen at the bean drop.

Live guidance

A dotted 60 s projection of where the bean temperature is heading, an FC countdown, and automatic **RoR crash / flick** warnings appear as the roast progresses.

Fan override

Any manual fan change during a profile pauses the profile's fan while the temperature curve keeps running. `Resume profile fan` hands it back at the profile's values. `Resume — shift curve to my fan` hands it back but offsets the whole remaining fan curve through the speed you chose.

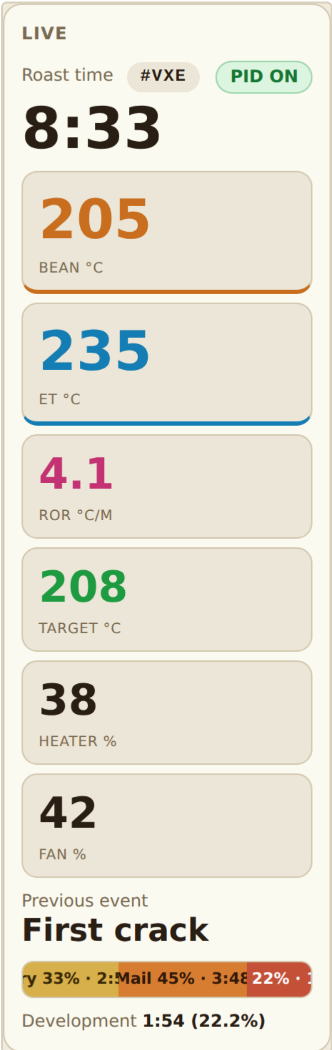


Fig. 13 — Late in the roast: First Crack is the previous event and the phase bar shows development.

Fig. 14 — The same rail in night mode.

8. After the roast: data, notes and comparisons

Everything is in the **Data** panel.

- Roasts are **saved automatically at Drop** with a date and time stamp and their batch code. Save this roast stores the current chart manually, for example after a test run. The drop-down lists saved roasts for Load and Del.
- **Roast notes** follows the current roast (the one roasting, just dropped, or loaded from the list) and saves as you type. ☆ Star marks an excellent roast so it stands out in the list.
- The bean and green weight you chose in the Start cycle dialog are stored with each roast and shown in its list entry.
- Download CSV exports the current chart's samples (time, bean and exhaust temperatures, heater, fan, target, events) with the roast details in a header, for Excel or Artisan.
- Overlay as background ghosts a saved roast onto the chart, aligned at Charge, and shows a live **BT delta vs background** readout so you can roast to match it. ✕ clear BG removes it.

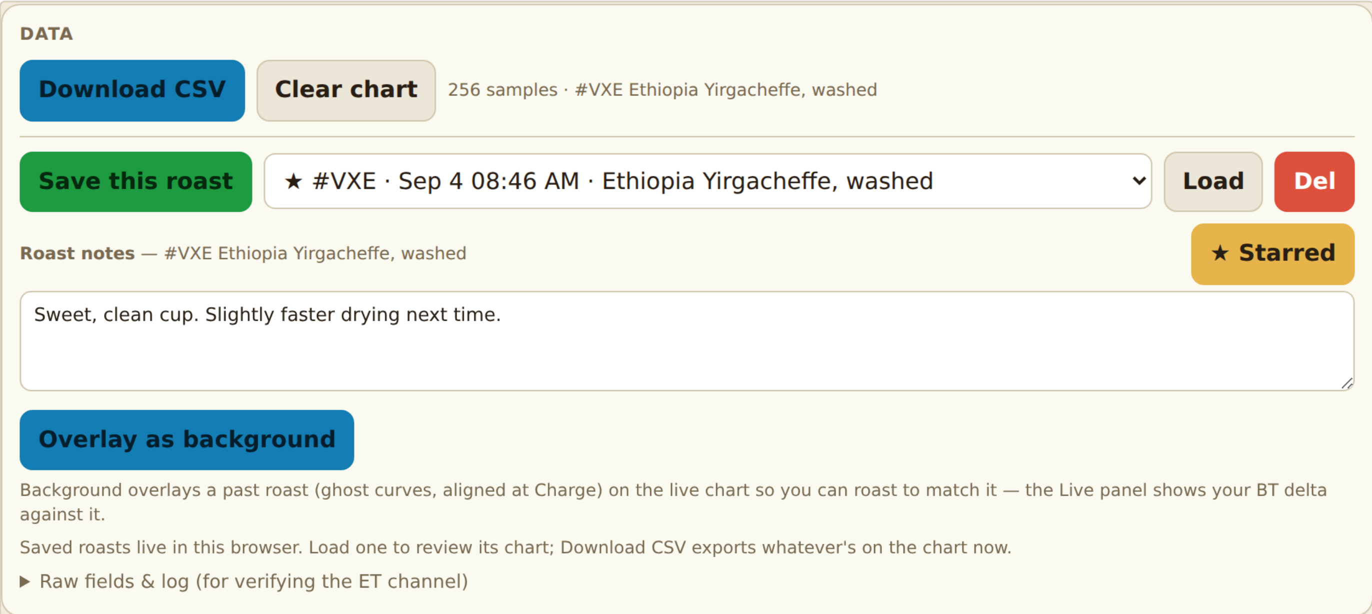


Fig. 15 — The Data panel with a starred roast loaded and a note typed in.

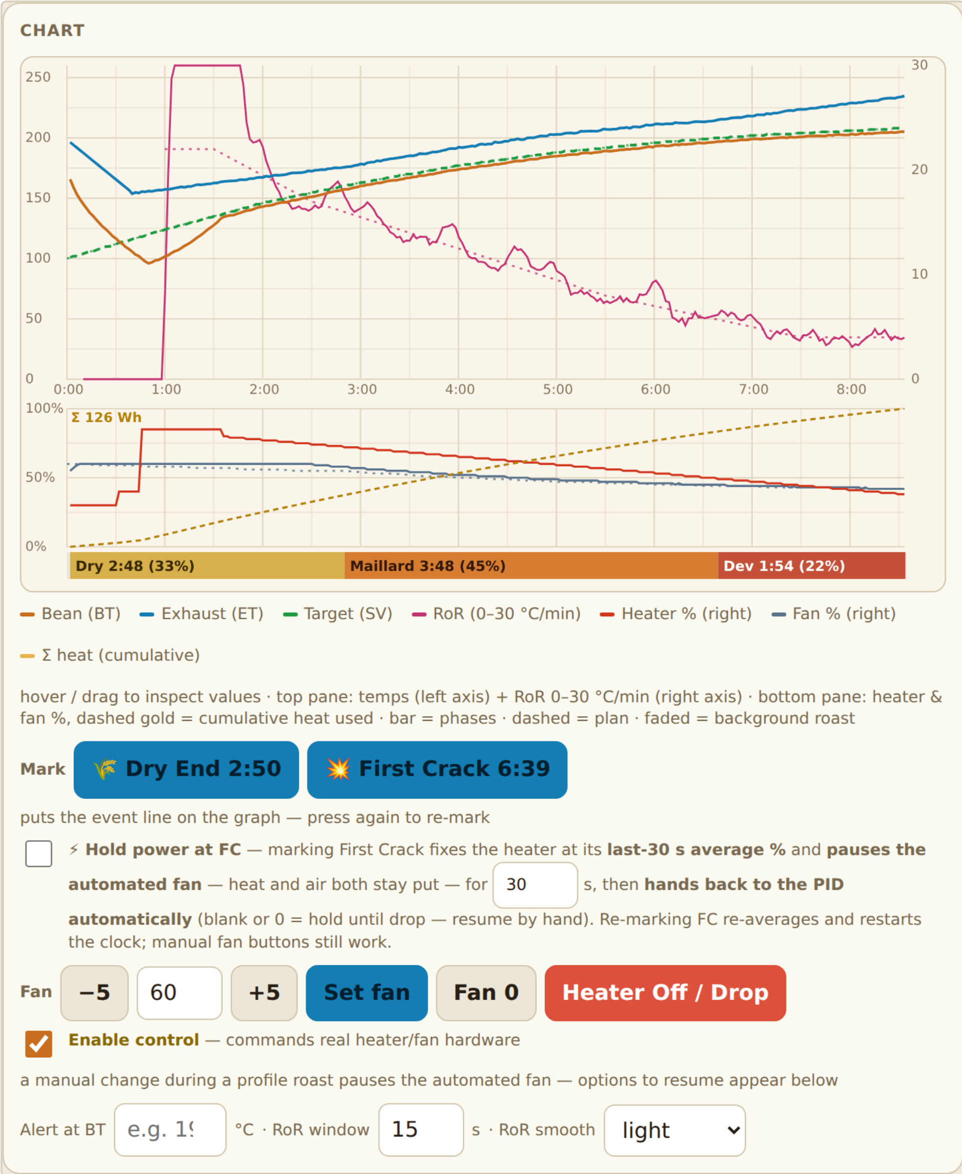


Fig. 16 — A saved roast loaded for review. Use Overlay as background to ghost it behind your next live roast.

9. Green coffee inventory

Optional, but it makes batch planning and stock tracking automatic.

1. In **Green coffee inventory** press + Add bean and enter the name, origin, available weight and notes. Or press 📷 Photo label (OCR) and photograph the bag's spec label. On-device text recognition (English and Chinese) fills the fields for you to check.
2. Set your roaster's **Batch size** range (for example 100–200 g). Each bean card then shows a suggested even split, such as *5 × 170 g*, so you do not end up with an awkward leftover. ⚖️ **Batch planner** does the same for any weight you type in.
3. When starting a roast, pick the bean and the green weight in the dialog that opens from Start between-batch cycle.
4. At Heater Off / Drop the batch's green weight is deducted from stock. Out-of-stock beans are flagged in the list.

Fig. 17 — Two beans in stock, each with its suggested batch split.

Fig. 18 — Adding a bean by hand.

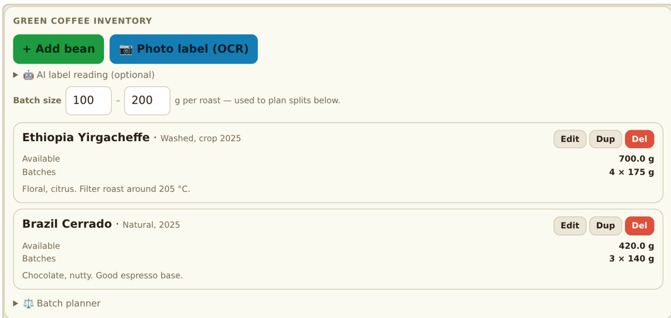


Fig. 19 — After the roast: 150 g deducted from the Ethiopia, and the split re-planned.

▼ ⚖️ Batch planner

Green to split | Ethiopia Yirgacheffe · 850 g | 850

✅ **5 × 170 g** — even split, nothing left over.
vs max-size: 4 × 200 g + **50 g** left over (too small to roast).
Other even splits: 6 × 141.7 g · 7 × 121.4 g · 8 × 106.3 g

Fig. 20 — The batch planner splits any weight into equal batches within your size range.

AI label reading (optional)

Inside 🤖 **AI label reading** you can point the photo reader at Claude or Gemini for much better results on messy or multi-language labels. Paste **your own** API key (stored in this browser only and never synced), or set a **Proxy Worker URL** so the key never lives on the device. The photo is sent to your own account and billed to you. Leave both fields empty to keep using on-device OCR.

▼ 🤖 AI label reading (optional)

Provider | Claude (Anthropic)

API key | paste your key (stored in this browser only)

Proxy Worker URL | optional — https://…workers.dev (keeps the key off this device)

With a key set, 📷 **Photo label** sends the photo to your chosen AI to read the label — far better on messy, dense, or multi-language labels. It fills the name, origin, bag weight, and a tidy spec summary. The photo is sent to **your own account** (billed to you). Get a key at **console.anthropic.com** (Claude) or **aistudio.google.com** (Gemini — has a free tier). Leave everything blank to use on-device OCR instead.

🔒 **Proxy Worker URL** (optional): route through your own Cloudflare Worker (deploy **ai-proxy-worker.js** from the repo) so the key lives on the Worker, not this browser — best for shared/public use. When a proxy is set the API-key field above is ignored. The key is kept only in this browser and never cloud-synced.

Fig. 21 — Provider, API key and optional proxy Worker URL.

10. Using more than one device (Cloud sync)

Profiles, roast logs and inventory can be shared between your phone, tablet and computer through a small Cloudflare Worker that **you** own. It is free and needs no account in the app.

Setup takes about 10 minutes once and is walked through click by click in `CLOUD_SYNC.md` in the repository. In short:

1. Create a free Cloudflare Worker from `sync-worker.js` with a KV namespace bound as `SYNC`.
2. Paste the Worker's URL into the **Cloud sync** panel.
3. Press Generate for a sync code, then Connect. Your local data is uploaded.
4. On every other device enter the same URL and code and press Connect. From then on syncing is automatic: a change pushes about 1.5 s after you make it, each device pulls every 45 s and whenever you switch back to the tab.

Fig. 22 — The Cloud sync panel.

> The sync code is both the address and the password. Anyone with it can read and write your data, so keep it private. Do not share your Worker URL publicly either.

11. Advanced mode (experienced users only)

Ticking **Advanced mode** below the Profile designer reveals two open-loop tools that drive the heater **directly in manual mode**. Stay at the machine while they run. Heater Off / Drop still works.

- **ET-track roast** — replays a saved roast by its *exhaust* temperature, using the reference's recorded heater % as feedforward with trim on the ET error, and replays its fan curve. The run ends when the bean temperature reaches the **Drop at BT** you set. Arm it for auto-charge to use it in the between-batch flow instead of a profile.
- **Power profile** — distils a saved roast's heater curve into a fixed schedule of power steps (one `time heater% [fan%]` per line) and runs it with zero mid-roast adjustments. Set **Drop at BT** as a safety net.

Both are experimental. Do a few normal profile roasts first so you know how your machine behaves.

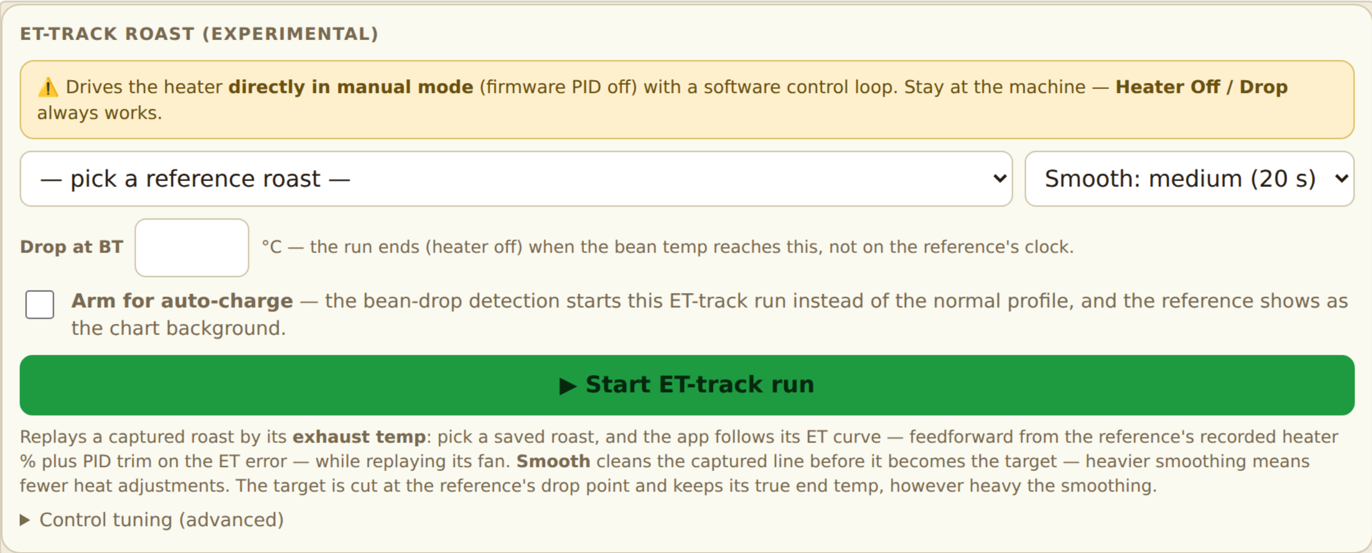


Fig. 23 — ET-track roast.

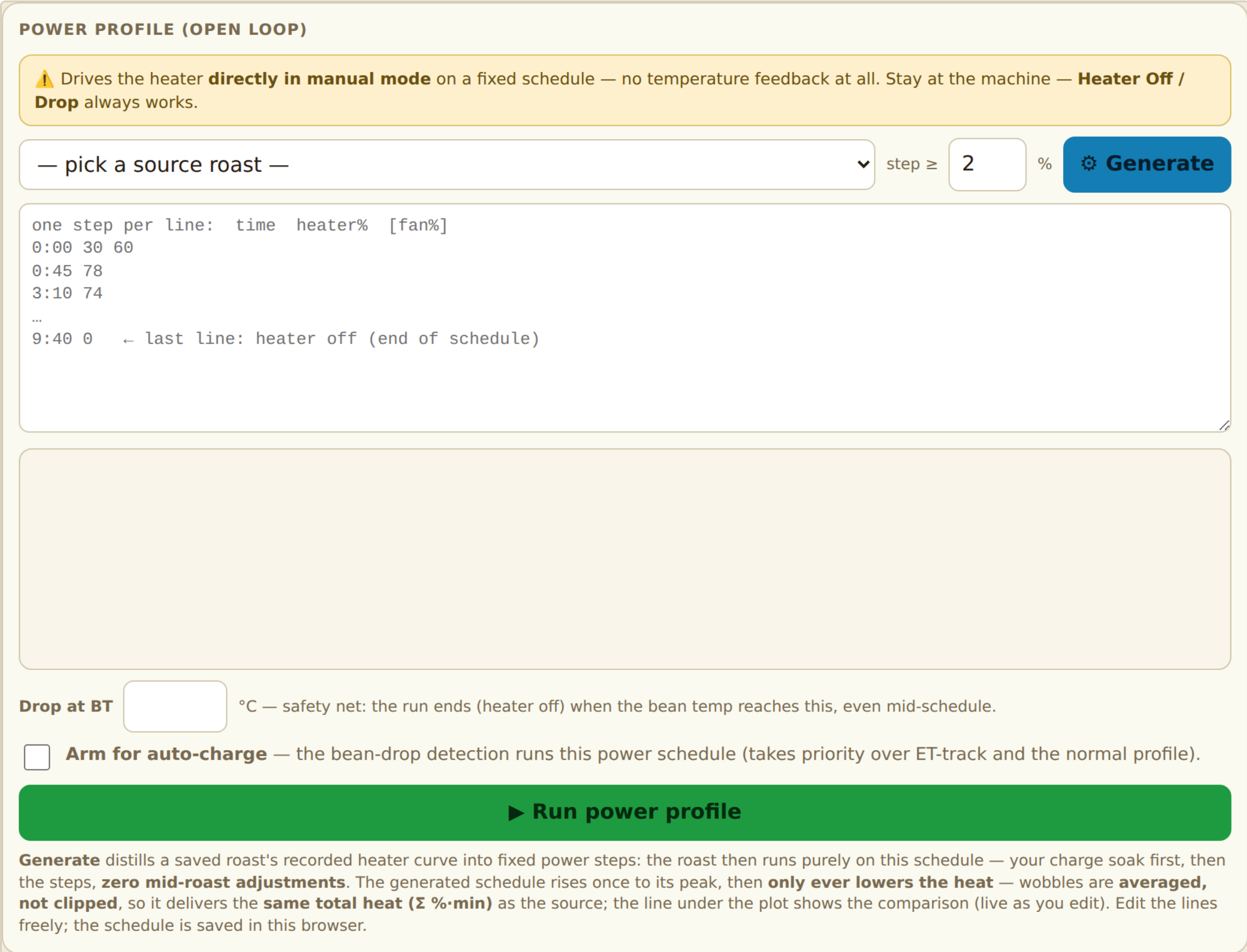


Fig. 24 — Power profile (open loop).

12. Safety checklist

- Fan on before heater on. The app enforces airflow in its automated flows, but in manual fan control it is on you.
- **Enable control** is per session. It resets when you reload the page, which is intentional.
- Stay with the machine. Alarms and warnings are aids, not supervision.
- Losing the Bluetooth link stops every automated run and the app shows **disconnected**, but the roaster's own state persists. If a roast is in progress when the link drops, reconnect immediately or turn the roaster's heater off at the machine.
- If in doubt: Heater Off / Drop.

13. Troubleshooting & FAQ

Problem	What to try
"Web Bluetooth isn't available" banner	Use Chrome or Edge, not Safari or Firefox. On iOS use another device.
Roaster not in the device list	Power-cycle the roaster, close the vendor app, tick **List all Bluetooth devices**, move closer.
Connected but tiles show –	Open **Connection details / advanced**. If the Notify/Write fields are –, the service was not found: use the free **nRF Connect** app to read the roaster's Service UUID, paste it into the custom Service UUID box, then reconnect.
Buttons do nothing	**Enable control** is not ticked, or the roaster is disconnected.
Heater will not fire	The fan is at 0. Set a fan speed first.
Auto-charge did not trigger	The bean-probe drop was too small or slow (small batch, low charge temperature). Press ▶ Run profile to start the curve now. Next time use a larger batch or a higher charge temperature so the plunge is clearer.
RoR trace is jagged	Increase **RoR smooth** or lengthen **RoR window** under the chart.
Profile started but the fan stayed put	No fan values in the profile and Auto fan off. Control the fan by hand, or tick **Auto fan** and re-run.
My saved data is gone	Data is stored per browser and per site URL. Check you are on the same URL and browser, or set up Cloud sync. Clearing site data deletes it.
I want °F	Not supported yet. The app works in °C throughout.

Still stuck? Open **Raw fields & log** at the bottom of the Data panel. It shows the raw telemetry frames and a log of every command sent, which is the first thing to include when asking for help.